

Stock Market Performance and Macroeconomic Factors: The Study of Indian Equity Market

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Vanita Tripathi

Ritika Seth

Abstract

There has been an extensive debate on the relationship between real economy and stock market performance especially in the context of emerging markets. This article examines the causal relationships between the stock market performance and select macroeconomic variables in India, using monthly data from July 1997 to June 2011. We use factor analysis, ADF and PP Unit root tests, Regression, ARCH model, Granger causality and Johansen Co-integration test for data analysis. Impulse Response analysis has also been performed to check the response of stock market to shocks created in the real economy.

We find a significant correlation among stock market indicators and macroeconomic factors. We identified three principal factors through Factor analysis viz Inflation, Interest rate and Exchange rate. The overall explanatory power of the regression model is 23.8%, 23.3% and 16.9% respectively for Sensex, Market capitalization and Market Turnover. There is uni directional causality from stock market to real economy. We find five co-integrating relationships between stock market and macroeconomic variables. These results suggest that the stock prices movement is not only the result of behaviour of key macroeconomic variables but it is also one of the important reasons of movement in other macro dimension in the economy.

Keywords

Stock market performance, Indian stock market, macroeconomic variables, unit root tests, Granger causality test, Johansen's co-integration test

Introduction

Stock market plays a crucial role in any country's economic growth and development. A healthy and flourishing stock market has been considered relevant for the nation's economic growth by channelizing capital towards investors and entrepreneurs. An economy is said to be efficient if it has a good banking system and a good stock market exhibiting upward trend (Kumari, 2011). There are two vital roles that

Vanita Tripathi, Assistant Professor, Department of Commerce, Delhi School of Economics, University of Delhi, India. E-mail: vanitatripathi1@yahoo.co.in

Ritika Seth, Research Scholar, Department of Commerce, Delhi School of Economics, University of Delhi, India. E-mail: ritika.pisces@yahoo.co.in

stock exchanges are required to carry out. First, they help in price discovery and, second, they provide liquidity. They provide a boost to the primary issues market and make it possible for corporate to undertake huge projects. A well performing stock exchange is very helpful for economic activity through growth and saving, efficient allocation of investment and attracting foreign direct investment (FDI). However, to perform this important task it is very necessary that stock market has a significant relationship with the macroeconomics variables.

Ross (1976) came out with Arbitrage Pricing Theory (APT), for estimating the asset's expected returns. APT essentially seeks to measure the risk premium attached to various risk factors and attempts to assess whether they are significant and if they are priced into stock market returns. By employing factor analysis, he asserted that there are several systematic factors (industry specific and macro-economic) that affect the security returns besides a security's beta, that is, the sensitivity of the individual security to the changes in the market return, such as gross domestic product, inflation and the structure of interest rates and so on, which could affect the firms differently.

Earlier a country was considered strong and efficient if it exhibited a sustained growth of Gross Domestic Product (GDP) and per capita income. But, of late it has been recognized that stock market exerts greater influence on national economy. Market capitalization, savings, investment, performance of stock market, consumption and sound banking and insurance system are considered to be a few important indicators of economic growth (Kumari, 2011). Stock markets and real economy are the two sides of the same coin or we can say that these two are like complementary goods which cannot be called complete without each other and cannot achieve the basic purpose of overall development of a country without each other.

The unusual rise and fall in of Bombay Stock Exchange (BSE) Sensitive Index (SENSEX) has received a lot of media attention over the last couple of decades in India. Even some policy analyst has designated it as an 'indicator' of India's inevitable growth and development. A persistent question that is raised in the debate of Indian stock markets is whether they are casinos where fortunes are made and lost at the throw of the dice or indeed they represent mature and meaningful institutions that are purely/ integrally related to the overall macro-economy. Do the stock exchanges represent institutions that affect crucial macro-economic variables or are they affected by these and hence serve as some kind of signals? (Pethe and Karnik, 2000) Therefore, whether stock markets lead or lag real economic activity is an empirical question.

Macroeconomic factors influence the stock market performance and particularly stock returns through their effect on future cash flows and the rate at which they are discounted. The six macroeconomic factors considered in the present study with their relationship with stock market are briefly explained in Table 1.

Table 1. Expected Theoretical Relationship between Stock Market and Economic Indicators

	Sensex Returns	Market Capitalization	Market Turnover
Inflation	Negative	Negative	Negative
Interest rate	Negative	Negative	Negative
Money supply	Negative/positive	Negative/positive	Positive
Oil prices	Negative	Negative	Negative/positive
Exchange rate	Positive/negative	Positive/negative	Negative
IIP	Positive	Positive	Positive

Source: Authors' computation.

In this research work, we explore the relation especially the short- and long-run causal relation between stock market performance and six important macroeconomic variables using monthly data by means of descriptive statistics, correlation analysis, factor analysis, regression analysis, some sophisticated statistical techniques such as ADF and PP unit root test, ARCH test, Granger causality test, bi-variate and multi-variate Johansen co-integration test and Impulse Response Analysis.

The rest of the article is arranged as follows—in the second section we present a brief review of existing literature related to the subject. The third section contains the objectives and the rationale behind the present study. The fourth section describes the data sources and the methodology used. The fifth section discusses the econometric results derived out of the application of several techniques. Finally, the sixth section provides the conclusion and implications of research findings.

Literature Review

Chen et al. (1986) were the first to employ specific macroeconomic factors as proxies for undefined variables in the APT. They examined equity returns relative to a set of macroeconomic variables and found that the set of macroeconomic variables which can significantly explain stock returns includes growth in industrial production, changes in the risk premium, twists in the yield curve, measures of unanticipated inflation and changes in expected inflation during periods of volatile inflation.

Mukherjee and Naka (1995) tried to analyze the dynamic relation between six macroeconomic variables and Japanese stock market returns and found that the cointegration relation existed and positive relationship was found between the Japanese industrial production and stock return. Abdalla and Murinde (1997), Nath and Samanta (2004) and Narayan (2009) investigate the interactions between exchange rates and stock prices. While Abdalla and Murinde did the research for India, Korea, Pakistan and the Philippines and reported unidirectional causality from exchange rates to stock prices in all countries except the Philippines, Nath examines the extent of integration between foreign exchange and stock market in India during the liberalization era using Granger's causality test in VAR framework and Gweke's Feedback Measures and found a very poor causal link between returns in foreign exchange and capital markets as per the results of Granger causality test. However, Gweke's Feedback Measures reported a high degree of integration between the two and even bi-directional as well as contemporaneous causal relationship between them. Narayan research revealed a positive relationship between exchange rate and stock returns. Also, appreciation of the Indian rupee reduces volatility more than depreciation increases the volatility.

Muradoglu et al. (2000), Kandir (2008) and Mohammad et al. (2009) investigate the role of macroeconomic factors, namely inflation, industrial production index, interest rate and foreign exchange rate in explaining the stock returns. Foreign exchange rate and interest rate appeared to be significant in explaining the stock returns, while other variables were not much significant. Pethe and Karnik (2000) study the relationship between stock market behaviour and some macroeconomic variables and found a weak causality running from IIP to Sensex (Nifty). Maysami et al. (2004) and Ahmed and Imam (2007) investigate the relationship between stock market and different macroeconomic variables in Singapore and Bangladesh. It was found that the Singapore stock market and the SES All-S Equities Property Index formed significant relationships with all macroeconomic variables identified. On the other hand, Bangladesh stock market does not reflect macroeconomic effect on stock price indices.

Agarwalla and Tuteja (2008) examine the relationship between share price index and economic growth for India by using multi-variate Granger causality test within an error correction framework and found the causality running from economic growth proxied by industrial production to share price index and not the other way round which shows that stock markets in India are still demand driven and industry led. Gay (2008) evaluated the association among stock prices and macroeconomic variables in cases of Brazil, Russia, India and China which are emerging economies of the world using oil prices and exchange rate as explanatory variables employing Box Jenkins ARIMA model and found insignificant results which postulate inefficiency in market. Singh (2010) made an attempt to explore the causal relation between stock market index, that is, BSE Sensex and macroeconomic variables of Indian economy using Granger causality test. He found that the IIP is the only variable having bilateral causal relationship with BSE Sensex. WPI is having strong correlation with Sensex, but it is having unilateral causality with BSE Sensex.

Basu and Chawla (2010) tested the validity of fundamental CAPM theory in India. It verifies the efficiency and efficacy of the model by examining 10 portfolios covering 50 stocks which are part of S&P CNX Nifty Index over a 5 year period from 1 January 2003 to 1 February 2008 using weekly data. The study employs regression analysis and finds that CAPM performs below expectations and it completely fails in the Indian context.

Basu and Chawla (2012) also tested the validity of the APT model in India. This article examines 10 portfolios covering 50 stocks which are part of S&P CNX Nifty from 1 January 2003 to 1 February 2008 using weekly data to verify the efficiency and efficacy of the model. The macroeconomic variables taken up in the study include 91 days treasury bill rate which is a proxy for the risk free interest rate, 10 year government security rate, the INR USD exchange rate, Wholesale Price Index (WPI) and gold prices. To overcome the problem of multicollinearity among the macroeconomic regressors, a factor analysis was carried out that resulted in two factors namely inflation and market index. The regression results display accurate relationships that are significant for each of the 10 portfolios and moderate to high explanatory power. Thus, it concludes that APT is a good fit in India over the chosen sample period.

Kumari (2011) investigates the relationship between stock returns and inflation in India and found that there is no significant relation between stock returns and inflation in the post-reform period in India. It implies that stock returns do not provide a hedge against inflation. Mishra and Singh (2011) test whether the stock market in India is driven by macroeconomic fundamentals. By using OLS and semi-parametric Generalized Additive Model (GAM), they found that industrial production and FII have positively affected volatility of stock returns. Moreover, macroeconomic variables explain more of Sensex than S&P CNX Nifty volatility.

Objectives and Rationale of Study

The objective of the present research work is to explore the relation especially the short- and long-run causal relation between stock market performance and six important macroeconomic variables using monthly data by applying a variety of statistical and sophisticated econometric techniques.

No previous study in the related subject had tried to find out the relationship between overall stock market performance and real economy. The previous literature has focussed only on finding the relationship between stock market returns and macroeconomic variables. While our study is one step forward in the sense that we have taken the overall stock market performance proxied by not just stock

returns but total market capitalization and total market turnover as well as the dependent variables. Also, we have applied Impulse Response Analysis to find out how responsive the stock market indicators are as to one standard deviation of shock created in the real economy and the vice versa with the help of graphs. The use of this technique is one of the uniqueness of our study as no previous study on the related subject has applied this technique.

The rationale behind the study is to check the informational efficiency of the Indian stock market. If lagged changes in some macroeconomic variables cause variations in stock prices and past fluctuations in stock prices cause variations in the economic variable, then bi-directional causality is implied between the two series. This behaviour indicates stock market inefficiency. If market is inefficient with respect to information, then it has important implications both at micro and macro levels. At the micro level, this implies that the individual investor can earn considerably higher normal rate of returns from the stock market. At the macro level, it raises serious doubts on the ability of the market to perform its fundamental role of channelling funds to the most productive sectors of the economy. In contrast, if changes in the economic variable neither influence nor are influenced by stock price fluctuations, then the two series are independent of each other and the market is informationally efficient.

Data and Methodology

The data consist of monthly time series observations regarding the macroeconomic variables namely, exchange rate, Index of Industrial Production (IIP), interest rate, money supply, oil prices and WPI and stock market indicators, namely BSE India Sensex, BSE India market capitalization and BSE India market turnover covering the 14 year period from July 1997 to June 2011. The data for macroeconomic variables such as inflation proxied by WPI have been taken from the official website of Ministry of Commerce and Industry (Office of the Economic advisor to the government of India); the data for real economic activity proxied by IIP have been collected from CMIE (Centre for Monitoring Indian Economy (Private) Limited) monthly publication, called, '*Economic Intelligence Service—Monthly Review of the Indian Economy*'. The data for interest rate proxied by implied yield on 91-days T-Bills are extracted from *Handbook of Statistics on Indian Economy* maintained by RBI. www.indexmundi.com is used for getting data relating to crude oil price (petroleum). Stock prices are represented by the monthly closing values of Index (SENSEX) taken from www.bseindia.com. The data for market capitalization, market turnover, money supply and exchange rate are taken from www.rbi.org.in. For the analysis of data and to arrive at meaningful conclusions, we have performed the whole analysis by taking Log of all the variables because log series do not assume a linear relationship among the variables. Also, Log Sensex in its differenced form represents the return on Sensex (Table 2).

Besides the usual trend analysis, descriptive statistics, correlation and regression analysis, we use the following statistical and econometric techniques.

Unit Root Test

Augmented Dickey Fuller (ADF) test and the Phillips and Perron (PP) test are conducted to examine whether the time series are stationary or not. A stochastic process is said to be stationary if its mean and variance are constant over time and the value of the co-variance between the two time periods depends

Table 2. Operational Definitions

Category	Variables Used	Definition	Symbol Used
Stock market performance	BSE Sensex	It is an index comprising of 30 stocks with base year 1978–1979 (month end adjusted value)	BSE
	BSE India market capitalization	It is the aggregate value of all the public stocks traded on the Bombay stock exchange (month end value in crores of rupees)	Marcap
	BSE India market turnover	It is the total number of shares traded multiplied by their respective market price at a certain point in time or within a certain period (in crores of rupees)	Marturn
Macroeconomic variables	Interest rate (91 days T-Bills rate)	A treasury bill is a promissory note issued by the government under discount for a specified period stated therein. Its implied yield in percentage is considered	Int
	WPI (wholesale price index)	It represents an overall upward price movement of goods and services. Inflation happens either when prices go up or when it takes more money to buy the same items (index with base year—1993–1994)	Inf
	Money supply	Broad money, i.e., M3 is used (actual value in crores of rupees)	MSupply
	Index of industrial production	It is an economic indicator which measures real production output, which includes manufacturing, mining and utilities (index with base year—1993–1994)	IIP
	Crude oil price (\$ per barrel)	Actual value of crude oil (petroleum) price (end of the month)	Oil
	Exchange rate (₹ per \$)	Actual value (end of the month)	Exchrates

Source: Authors' computation.

only on the distance or gap or lag between the two time periods and not the actual time at which the co-variance is computed. It is very important that a time series must be stationary because if a time series is non-stationary, we can study its behaviour only for the time period under consideration. As a consequence, it is not possible to generalize it to other time periods. Therefore, for the purpose of forecasting, such (non-stationary) time series may be of little practical value.

The equation used for conducting ADF test for a series X_t has the following general structure:

$$X_t = \alpha + \beta t + \rho X_{t-1} + \sum_{k=1}^n \delta_k \Delta X_{t-k} + e_t \quad (1)$$

In the above equation, if

1. $\beta = 0$ and $|\rho| < 1$, then series X_t is stationary.
2. $\beta = 0$ and $\rho = 1$, then the series is an $I(1)$ process.
3. $\beta \neq 0$ and $|\rho| < 1$, then the series is trend stationary around a deterministic time trend.

e_t = pure white noise error term.

Phillips and Perron have developed a more comprehensive theory of unit root non-stationarity. The tests are similar to ADF tests, but they incorporate an automatic correction to the DF procedure to allow for auto correlated residuals. They use non-parametric statistical methods to take care of the serial correlation in the error terms without adding lagged difference terms. The tests often give the same conclusions and suffer from most of the same important limitations as the ADF tests.

Factor Analysis

Factor analysis has been applied on the explanatory variables to arrive at a few factors and then use these factors as independent variables for estimating the regression equation. It is a statistical method used to describe variability among observed, correlated variables in terms of a potentially lower number of unobserved variables called factors. It attempts to identify the underlying structure in a data set by defining a small number of factors that capture the variation in the collected data. It can help to identify the underlying causal structure in the data set. It can also help in the further analysis of the data by reducing the number of variables and simplified model selection.

ARCH Model

AutoRegressive Conditional Heteroskedasticity (ARCH) models are used to characterize and model observed time series with time-varying volatility such as stock prices, exchange rates and inflation rates. ARCH models assume that the variance of the current error term is related to the size of the previous periods' squared error terms, giving rise to volatility clustering. This phenomenon is widely observable in financial markets, where periods of low volatility are followed by periods of high volatility and vice versa. The ARCH model can be seen as:

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \dots + \alpha_q \epsilon_{t-q}^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2 \quad (2)$$

where $\alpha_0 > 0$ and $\alpha_i \geq 0, i > 0$. All things being equal, α_i carries more intense influence as compared to α_j . That is, older news holds less impact on current investment decisions, which results in comparatively lesser volatility than the current news.

Granger Causality Test

In order to check the short-term causal relationship among stock market indicators and macroeconomic variables, Granger causality test is used. This test states that if past values of a variable X significantly contribute to forecast the value of another variable Y , then X is said to Granger cause Y and vice versa (Granger 1969, 1988).

The test is based on the following two regressions equations. These equations can be used to show the direction of causality among stock market indicators and macroeconomic variables.

$$X_t = \alpha_1 + \sum_{k=1}^m \beta_{1k} X_{t-k} + \sum_{k=1}^m \gamma_{1k} Y_{t-k} + e_{1t} \quad (3)$$

$$Y_t = \alpha_2 + \sum_{k=1}^m \beta_{2k} Y_{t-k} + \sum_{k=1}^m \gamma_{2k} X_{t-k} + e_{2t} \quad (4)$$

where Y_t and X_t are the variables to be tested, e_{1t} and e_{2t} are mutually uncorrelated white noise errors, t denotes the time period, k is the number of lags and m is the maximum number of lagged observations included in the model. The null hypothesis is $\gamma_1 = \gamma_2 = 0$ for all k 's versus $\gamma_1 \neq \gamma_2 \neq 0$ for at least some k 's. If the coefficients γ_1 are significantly significant but γ_2 are not, then Y granger causes X . However, in the reverse case, X granger causes Y . And, if both γ_1 and γ_2 are significant then causality runs in both directions. Finally, if both γ_1 and γ_2 are not statistically different from zero, then X and Y will be independent.

Johansen Co-integration Test

Using non-stationary series, cointegration analysis has been used to examine whether there is any long-run equilibrium relationship. When two variables are cointegrated, it implies that the two time series cannot wander off in opposite directions for very long without coming back to a mean distance eventually. But it does not mean that on a daily basis the two series have to move in synchrony at all. Cointegration means that despite being individually non-stationary, a linear combination of two or more time series can be stationary. Cointegration analysis becomes important for the estimation of error correction models (ECM). The concept of error correction refers to the adjustment process between short-run disequilibrium and a desired long-run position.

The Johansen method applies the maximum likelihood procedure to determine the presence of cointegrating vectors in non-stationary time series as a vector autoregressive (VAR). Consider a VAR of order k

$$Y_t = A_1 Y_{t-1} + A_2 Y_{t-2} + \dots + A_k Y_{t-k} + BX_t + \varepsilon_t \quad (5)$$

where Y_t is a k -vector of non-stationary $I(1)$ variables, X_t is a the vector of deterministic variables, k is the number of lags and ε_t is a vector of innovations or error terms. We can rewrite the VAR as:

$$\Delta Y_t = \pi Y_{t-1} + \sum_{i=1}^{k-1} \tau_i \Delta Y_{t-i} + BX_t + \varepsilon_t \quad (6)$$

where

$$\pi = \left(\sum_{i=1}^k A_i \right) - I \quad (7)$$

$$\tau_i = - \sum_{j=i+1}^k A_j \quad (8)$$

Here Y_t is a vector of non-stationary variables. The information on the coefficient matrix between the levels of the series Π is decomposed as $\Pi = \alpha\beta'$ where the relevant elements of the α matrix are adjustment coefficients and the β matrix contains the cointegrating vectors. Johansen and Juselius (1990) specify two likelihood ratio test statistics to test for the number of cointegrating vectors. The first likelihood ratio statistics for the null of exactly r cointegrating vectors against the alternative of $r + 1$ vectors is the maximum eigenvalue statistic. The second statistic for the hypothesis of at most r cointegrating vectors against the alternative is the trace statistic. The number of lags applied in the cointegration tests is based on the information provided by the multivariate generalization of the AIC.

Also, AIC (Akaike Information Criterion) and SIC (Schwarz Information Criterion) criterions have been used to determine the optimal lag length structure.

Impulse Response Analysis

As a univariate AR model can be represented as an infinite order MA process, similarly a stable VAR model of order k can be represented as an infinite order vector moving average (VMA) process. This representation serves as the key to study the impulse response function (IRF). An IRF allows one to trace the time path of the impact of a shock in one variable on all the variables included in the VAR model. However the standard impulse response analysis assumes that shock occurs in one variable at a time. Such an assumption may be valid only if the shocks in different variables are not correlated. When there is a significant contemporaneous correlation between the variables, the results of impulse response analysis depends on the method of decomposition used. Here we have used Cholesky decomposition method which is one of the popular methods for this purpose.

Impulse Response Analysis has been performed to check the response of stock market indicators to shocks created in the real economy with the help of graphs.

We have divided our total study period into two sub-periods with the break taking place in 2007 (precisely 9 August 2007) which was the onset of global financial crisis, so as to determine the changes if any, in the relationship between stock market performance and macroeconomic variables have occurred over 14 years (1997–2011).

Sub-period I: July 1997–July 2007 (121 months)

Sub-period II: August 2007–June 2011 (47 months)

Therefore, all the tests are applied on the total time period as well as on the sub-periods.

Discussion of Results

Trend Analysis

We study the trend of all the independent variables namely, inflation, interest rate, IIP, exchange rate, oil prices and money supply as well as the dependent variables, that is, stock returns represented by SENSEX, market capitalization and market turnover with the help of graphs presented in Figures 1–9 by using monthly data.

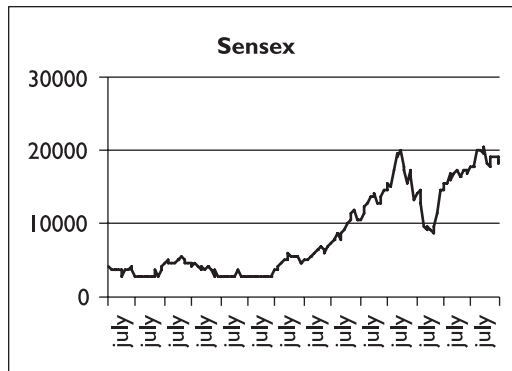


Figure 1. Trend of Sensex

Source: Authors' own.

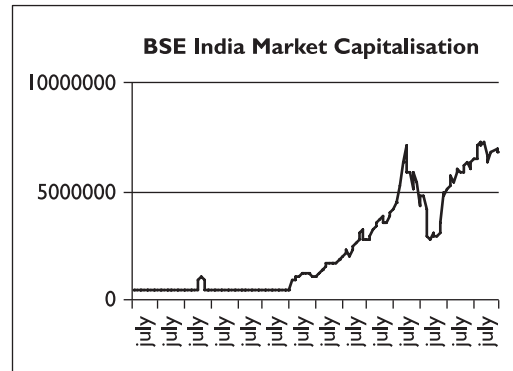


Figure 2. Trend of Market Capitalization

Source: Authors' own.



Figure 3. Trend of Market Turnover

Source: Authors' own.

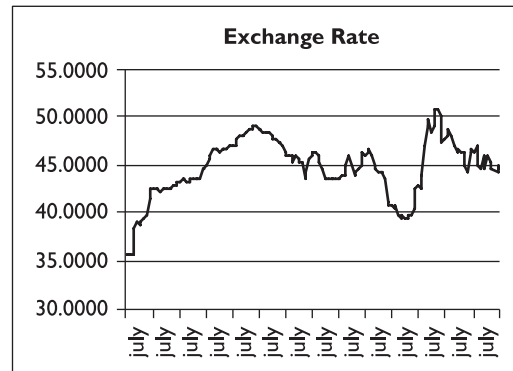


Figure 4. Trend of Exchange Rate

Source: Authors' own.

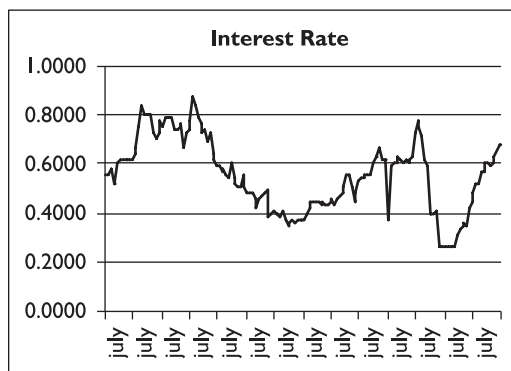


Figure 5. Trend of Interest Rate

Source: Authors' own.

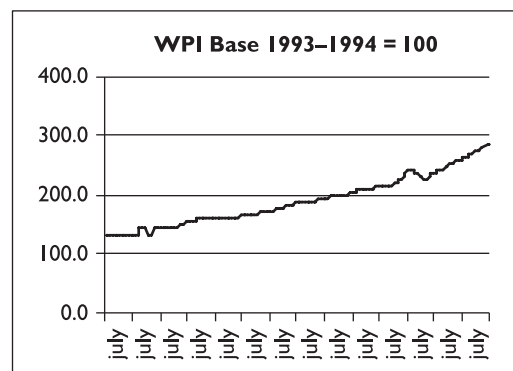
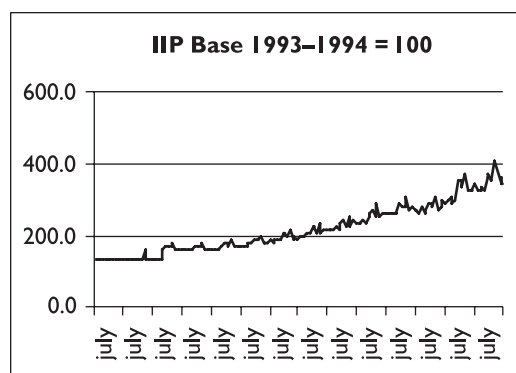
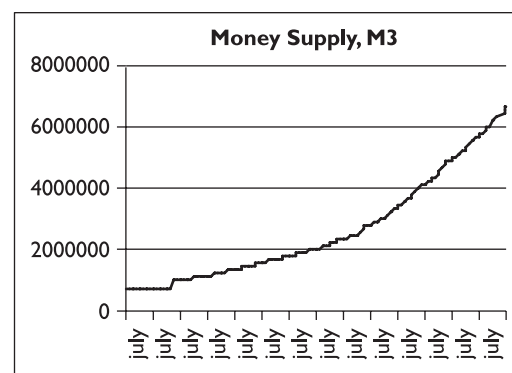


Figure 6. Trend of WPI

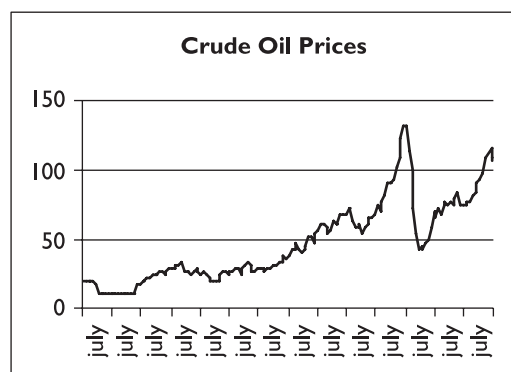
Source: Authors' own.

**Figure 7.** Trend of IIP

Source: Authors' own.

**Figure 8.** Trend of Money Supply

Source: Authors' own.

**Figure 9.** Trend of Oil Prices

Source: Authors' own.

Descriptive Statistics

From Table 3, it can be read that the mean value of Sensex is 8,622.531, while market turnover and market capitalization have mean values of ₹65,044.44 and ₹2,448,597, respectively. The coefficient of variation is highest in the case of market capitalization which indicates maximum volatility and it is lowest in the case of exchange rate.

From the skewness measure, we found that all variables except exchange rate and interest rate are positively skewed or skewed to the right (has a right tail) as their mean value is greater than the mode and median values. Interest rate has a skewness value of almost 0, so we can say that it is normally distributed.

Kurtosis values of all variables show that the data are not normally distributed because of kurtosis values are deviated from 3. All variables except exchange rate and market turnover have kurtosis less than three, so they are platykurtic or have a flat distribution. Exchange rate and market turnover have kurtosis value greater than 3, implying that their distribution is Leptokurtic or fat-tailed and sharply peaked about the mean when compared with the normal distribution.

Table 3. Descriptive Statistics (Level Variables)

Series	Sensex	Market Turnover (₹ crores)	Market Capitalization (₹ crores)	IIP	Oil Prices (\$ per Barrel)	Interest Rate (Implied Yield (%))	WPI	Exchange Rate (₹ per \$)	Money Supply (₹ crores)
Mean	8,622.531	65,044.44	2,448,597	217.9899	47.403	0.5504	190.4879	44.7756	2,644,465
Median	5,661.3	58,790.49	1,245,957	200.4500	37.780	0.557308	185.900	45.0250	2,062,911
Maximum	20,509.09	199,088.7	7,294,570	401.8601	132.550	0.8722	286.6692	50.9450	6,697,422
Minimum	2,810.66	15,326.00	446,728	131.5013	10.410	0.2627	131.600	35.7100	726,205
Std. deviation	5,737.199	38,623.74	2,213,556	65.90086	29.043	0.1461	41.3939	2.99422	1,687,917
Skewness	0.682899	0.790741	0.843444	0.666374	0.8123	0.0463	0.5194	-0.65203	0.861883
Kurtosis	1.920898	3.281794	2.232197	2.424340	2.8558	2.1956	2.27498	3.353274	2,547,156
Jarque-Bera	21.20906	18.06344	24.04578	14.75322	18.619	4.5888	11.23432	12.77788	22.23508
Jarque-Bera (probability)	0.000025	0.000120	0.000006	0.000626	0.000091	0.100821	0.003635	0.001680	0.000015
Coefficient of variation	0.665373	0.593805	0.90401	0.302312	0.612683	0.265443	0.217305	0.066872	0.638283

Source: Authors' computation.

Correlation Analysis

Table 4 shows that only interest rate and exchange rate are negatively related to the stock market variables, where only interest rate is significantly related. Other four independent variables are positively and significantly correlated to the stock market variables.

Similar results are found when the correlations are calculated among the variables in log form.

In Table 4, the correlation coefficients which are highlighted in bold and underlined shows the presence of *multicollinearity* among the explanatory variables, namely WPI & IIP, WPI & oil prices, WPI & money supply, IIP & oil prices, IIP & money supply and oil prices & money supply. Another way to detect and overcome the problem of multicollinearity is to carry out a factor analysis of the independent variables and arrive at a few factors and then use these factors as independent variables for estimating regression equation. It is known that these factors would be uncorrelated and thereby one would not face the problem of multicollinearity.

Factor Analysis

Factor analysis requires the number of observations to be at least 4–5 times the number of variables and in our research; there are 168 observations as against six explanatory variables. The Kaiser–Meyer–Olkin (KMO) statistic, which is a measure of sample adequacy, is 0.783. The KMO measure of sampling adequacy is an index for comparing the magnitudes of the observed correlation coefficients to the magnitudes of partial correlation coefficients. Large values for the KMO measure indicate that a factor analysis of the variables is a good idea. The KMO statistic ranges from 0 to 1. Another indicator of the strength of the relationship among variables is Bartlett's test of sphericity which requires that the correlation matrix of the variables used in the study should be significant. It tests whether the correlation matrix is an identity matrix, which would indicate that the factor model is inappropriate. In the present research Bartlett's test of sphericity converts into a chi-square value of 1,810 which is highly significant as the corresponding *p* value for this is 0.000.

Six explanatory variables were factor analyzed using principal component analysis with Varimax (orthogonal) rotation method. The principal component method helps in identifying the number of factors to be extracted while the varimax rotation method helps in better interpretation of the results of factor analysis. It was found that three factors accounting for about 98.086 per cent of the total variations were extracted. A cut-off point of 0.6 as factor loading was used for interpretation of the results.

The first factor with variables having a factor loading of 0.6 and above were WPI, IIP, oil prices and money supply. Therefore, this factor was labelled as Inflation Factor (INF). The second factor had only one variable, namely exchange rate, and the third factor also had only one variable, namely interest rate. The first factor accounted for 62.879 per cent of the variation, whereas the second and third factor accounted for 17.722 per cent and 17.485 per cent of the variation, respectively.

Sub-period Results—Correlation Analysis

The results of sub-periods are different from the entire study period. In sub-period I, exchange rate and interest rate are positively related to the stock market turnover. In sub-period II, all macroeconomic factors except exchange rate are positively correlated to stock market index and market capitalization. Exchange rate is significantly negatively correlated to stock market indicators. WPI, IIP, interest rate and

Table 4. Karl-Pearson's Correlation Matrix

	Wholesale Price Index (93-94 = 100)	Index of Industrial Production (93-94 = 100)	Exchange Rate (₹ per \$)	Crude Oil Price (\$ per barrel)	Interest Rate	Money Supply	BSE Sensex	Market Capitalization (in crores)	Market Turnover (in crores)
Wholesale price index (93-94 = 100)	1	0.977**	0.237**	0.892**	-0.366**	0.985**	0.909**	0.930**	0.627**
Index of industrial production (93-94 = 100)		0.000	0.002	0.000	0.000	0.000	0.000	0.000	0.000
Exchange rate (₹ per \$)		I	0.173*	0.869**	-0.346**	0.977**	0.928**	0.945**	0.654**
			0.025	0.000	0.000	0.000	0.000	0.000	0.000
			I	-0.033	-0.356**	0.214**	-0.082	-0.035	-0.070
Crude oil price (\$ per barrel)				0.668	0.000	0.005	0.290	0.657	0.370
Interest rate				I	-0.145	0.852**	0.904**	0.901**	0.716**
				0.061	0.000	0.000	0.000	0.000	0.000
Money supply				I	-0.349**	-0.192*	-0.217**	-0.217**	-0.046
					0.000	0.013	0.005	0.005	0.554
BSE Sensex					I	0.913**	0.943**	0.943**	0.621**
						0.000	0.000	0.000	0.000
Market capitalization (in crores)						I	0.991**	0.991**	0.783**
Market turnover (in crores)							0.000	0.000	0.000
							I	I	0.753**
									0.000
									I

Source: Authors' computation.

Notes: **Correlation is significant at the 0.01 level (2-tailed). *Correlation is significant at the 0.05 level (2-tailed).

money supply are negatively related to market turnover, while oil price is positively related. The result of multicollinearity also differs in sub-period II as compared to the entire study period. In sub-period II, there is multicollinearity among WPI & IIP, WPI & money supply, IIP & money supply and oil & interest rate.

Unit Root Test

The results of ADF and PP unit root tests on levels and at first differences for the entire period into consideration are presented in Tables 5 and 6. It is found that the null hypothesis, that is, time series has a unit root, could not be rejected for both stock market and macroeconomic variables at level series

Table 5. Results of Unit Root Test on Level Variables (Log)

Variable	On Levels			
	Constant & No Trend		Constant and Trend	
	ADF	PP	ADF	PP
LogSensex	-0.163843	-0.341197	-2.277219	-2.467869
LogMarcap	-0.020384	-0.145485	-2.242401	-2.446031
LogMarturn	-2.476035	-2.363936	-3.017110	-2.970456
LogInterest	-2.011711	-2.058340	-2.029121	-2.112361
LogWPI	0.821098	1.087578	-2.318993	-1.798789
LogIIP	0.656820	-0.262364	-2.180213	-7.621476*
LogExchrate	-3.245280**	-3.350800**	-2.980241	-3.002439
LogMsupply	0.730713	0.541189	-1.598673	-1.473222
LogOil	-0.944929	-0.905632	-3.299183***	-3.193362***

Source: Authors' computation.

Notes: *Significant at 1 per cent; **Significant at 5 per cent; ***Significant at 10 per cent.

Table 6. Results of Unit Root Test on First Difference of Log Variables

Variable	On First Difference			
	Constant & No Trend		Constant and Trend	
	ADF	PP	ADF	PP
LogSensex	-11.94562*	-11.99543*	-11.94233*	-11.99085*
LogMarcap	-11.36119*	-11.35179*	-11.34797*	-11.33775*
LogMarturn	-14.01262*	-14.22763*	-14.00305*	-14.24515*
LogInterest	-14.86206*	-14.73243*	-14.83594*	-14.73911*
LogWPI	-8.195462*	-8.207018*	-8.267822*	-8.281133*
LogIIP	-2.771884***	-38.40193*	-2.923641	-39.11840*
LogExchrate	-10.46758*	-10.38359*	-10.57409*	-10.52973*
LogMsupply	-2.447987	-12.81097*	-2.578545	-12.79010*
LogOil	-9.800165*	-9.776857*	-9.776928*	-9.753725*

Source: Authors' computation.

Notes: *Significant at 1 per cent; **Significant at 5 per cent; ***Significant at 10 per cent.

at 1 per cent level of significance. By taking the first order differencing for all these variables, the null hypothesis for non-stationarity was rejected for all variables at the same confidence level lending continuity in the modelling process.

Regression Analysis

Since the variables are non-stationary at levels, so their first differences are used for the application of regression test.

$$\text{Stock market performance} = f(\text{exchange rate, interest rate, WPI, oil prices, money supply, IIP})$$

The result of factor analysis has given us three main factors. Therefore, the independent variables in above regression model were replaced by these three factors namely inflation, exchange rate and interest rate and therefore the regression model was rewritten as:

$$\text{Stock market performance} = f(\text{inflation, exchange rate, interest rate})$$

We expect that the constant term will be statistically insignificant as the stock market performance is well explained by the three factors resulted through factor analysis. Further, we expect a negative relationship between inflation and stock market performance. An increase in inflation is expected to increase the nominal risk free rate which in turn will increase the discount rate used in valuating stocks. If cash flows also increase at the same rate, the effect of higher discount rate will be neutralized. However, the effect of a higher discount rate would not necessarily be neutralized by an increase in cash flows resulting from inflation, primarily because cash flows do not generally grow at the same rate as inflation. Therefore, if contracts are nominal and cannot adjust immediately, the effect will be negative on stock returns because cash flows would probably decrease initially if the cost of inputs adjusts faster to rising inflation than output prices. Further, inflation is perceived as negative news by the stock market, because it tends to curtail consumer spending and therefore company earnings. It also affects the value of the domestic currency adversely in foreign exchange market.

Since, by and large, inflation is negatively correlated with the stock prices; its relationship with total market capitalization would also be negative, as market capitalization is the product of stock prices and the number of shares outstanding. With a rise in inflation, the market capitalization would go down. Further, the relationship between inflation and total market turnover will also be negative. Because of inflation, stock prices are likely to reduce.

The relationship between exchange rate and stock market performance depends upon the nature of the firm. The sensitivity of an importing firm to a change in exchange rate is just opposite to that of an exporting firm.

Lastly, the relationship between interest rate and stock market performance is also expected to be negative because the lower the interest rate, the lower the cost of capital for the corporate sector as most of the companies finance their capital equipments and inventories through borrowings and therefore,

higher the corporate earnings, which should have a positive influence on the stock prices. Also, with a decline in interest rate, more and more people would like to diversify their portfolio holdings towards the stock market because of higher stock returns, leading to an increase in volume of trade as well as total market capitalization. Therefore, we can say that the relationship between interest rate and stock market indicators is negative.

The regression model was estimated using OLS method at 5 per cent level of significance. The results indicate that the factors have their a priori signs and the overall explanatory power of the model given by adjusted R^2 is 23.8 per cent, 23.3 per cent and 16.9 per cent respectively for Log Sensex, Log market capitalization and Log market turnover.

Sub-period Results

The regression results of sub-period I are quite similar to the entire study period but, in sub-period II, the explanatory power of the model is significantly increased. Therefore, we can say that the stock market has become more responsive to changes in domestic economy during the recent crisis period.

ARCH Model

ARCH model has been applied to check if our time series data suffers from the problem of heteroskedasticity. Heteroskedasticity refers to a situation when the standard deviations of a variable, monitored over a specific period of time, are non constant. It often arises in two forms, conditional and unconditional. Conditional heteroskedasticity identifies non-constant volatility when future periods of high and low volatility cannot be identified. Unconditional heteroskedasticity is used when future periods of high and low volatility can be identified. Conditional heteroskedasticity is often seen in the prices of stocks and bonds.

The possible existence of heteroskedasticity is a major concern in the application of regression analysis because the presence of heteroskedasticity can invalidate statistical tests of significance that assume that the modelling errors are uncorrelated and normally distributed and their variances do not vary with the effects being modelled.

Our null hypothesis is that the residuals are not heteroskedastic meaning residuals are homoskedastic as against the alternate hypothesis that the residuals are heteroskedastic.

We will check our regression model for heteroskedasticity using ARCH model. We have modelled stock market performance as a function of six macroeconomic variables. Log differenced series have been considered as all time series are stationary at first difference.

From Table 7, it is clear that we do not reject the null hypothesis of the residuals terms being homoskedastic in the case of Sensex and market capitalization as the dependent variable. Thus, our

Table 7. Test results for checking Heteroskedasticity

Dependent Variable	Obs* R-squared	Prob. Chi-square(1)
Sensex	0.431749	0.5111
Market capitalization	0.441888	0.5062
Market turnover	7.131505	0.0076

Source: Authors' computation.

regression model is free from the problem of heteroskedasticity. However, we reject the null hypothesis in the case of market turnover as the dependent variable.

Granger Causality Test

Before applying the Granger causality test, we have used AIC and SIC to ascertain the optimum lag length, which has come out to be two.

The results of Granger causality are summarized in Tables 8 and 9.

Table 10 reveals that much of the variation in economic variables is caused by stock market indicators and not the vice versa. While only exchange rate is granger causing Sensex returns and market capitalization, the other macroeconomic variables namely IIP, oil prices and WPI are granger caused by the fluctuations in Sensex returns and market capitalizations. Money supply is also granger causing market turnover. There is bi-directional causality between IIP and WPI. Further, WPI is granger causing interest rate and money supply and oil prices is granger causing IIP and interest rate.

Table 8. Granger Causality Table—2 Lags (Log Differenced Series)

	Dlogsensex	Dlogmarturn	Dlogmarcap	Dlogoil	Dlogwpi	Dlogiip
Dlogexchrte	→*	N	→*	→	N	←*
Dlogiip	←	N	←	←	↔	N
Dlogint	N	N	N	←	←	N
Dlogoil	←	N	←	N	N	N
Dlogwpi	←*	N	←	N	N	↔
Dlogmsupply	N	→	N	N	←	←

Source: Authors' computation.

Note: *Significant at 10 per cent.

Table 9. Granger Causality Table—2 Lag (Sub-Periods) Log Differenced Series

	Dlogsensex		Dlogmarturn		Dlogmarcap		Dlogoil		Dlogwpi		Dlogmsupply		Dlogiip	
	Sub-pd I	Sub-pd II	Sub-pd I	Sub-pd II	Sub-pd I	Sub-pd II	Sub-pd I	Sub-pd II	Sub-pd I	Sub-pd II	Sub-pd I	Sub-pd II	Sub-pd I	Sub-pd II
Dlogexchrte	N	N	N	N	N	N	N	→	N	N	N	N	←*	N
Dlogiip	←	N	→	←	←	N	N	N	↔	N	←	N	N	N
Dlogint	N	←	N	←	N	←	N	←	N	←	N	N	N	N
Dlogoil	↔*	←	N	N	N	←	N	N	N	↔	N	N	N	N
Dlogwpi	→*	N	→*	N	↔*	N	N	↔	N	N	N	N	↔	N
Dlogmsupply	N	N	N	N	N	N	N	N	N	←	N	N	→	↔*

Source: Authors' computation.

Notes: *Significant at 10 per cent.

Table 10. Results of Johansen's Co-integration Test

Hypotheses	Null	Trace Test	Maximum Eigen Value	Co-integrating Relationship	Number of Co-integrating Relationships
No co-integration between Sensex and exchange rate	$r = 0$ $r \leq 1$	9.119453 0.528201	8.591251 0.528201	No	0
No co-integration between Sensex and IIP	$r = 0$ $r \leq 1$	13.29890** 8.39E-07	13.29889** 8.39E-07	Yes	1
No co-integration between Sensex and interest rate	$r = 0$ $r \leq 1$	7.978364 1.732304	6.246061 1.732304	No	0
No co-integration between Sensex and money supply	$r = 0$ $r \leq 1$	53.49229* 7.619596*	45.87269* 7.619596*	Yes	2
No co-integration between Sensex and oil	$r = 0$ $r \leq 1$	23.90048* 0.348127	23.55236* 0.348127	Yes	1
No co-integration between Sensex and WPI	$r = 0$ $r \leq 1$	14.38168 4.777583*	9.604098 4.777583*	No	0
No co-integration between market turnover and exchange rate	$r = 0$ $r \leq 1$	14.71581 5.104635*	9.611177 5.104635*	No	0
No co-integration between market turnover and IIP	$r = 0$ $r \leq 1$	7.844321 0.001160	7.843161 0.001160	No	0
No co-integration between market turnover and interest rate	$r = 0$ $r \leq 1$	9.605270 3.585591**	6.019679 3.585591**	No	0
No co-integration between market turnover and money supply	$r = 0$ $r \leq 1$	52.76746* 4.642276*	48.12519* 4.642276*	Yes	2
No co-integration between market turnover and oil	$r = 0$ $r \leq 1$	18.98805* 3.820076*	15.16797* 3.820076*	Yes	1
No co-integration between market turnover and WPI	$r = 0$ $r \leq 1$	12.68230 1.195344	11.48695 1.195344	No	0
No co-integration between market capitalization and exchange rate	$r = 0$ $r \leq 1$	9.742879 0.126591	9.616287 0.126591	No	0
No co-integration between market capitalization and IIP	$r = 0$ $r \leq 1$	18.29368* 0.002234	18.29144* 0.002234	Yes	1
No co-integration between market capitalization and interest rate	$r = 0$ $r \leq 1$	7.731245 1.301657	6.429588 1.301657	No	0
No co-integration between market capitalization and money supply	$r = 0$ $r \leq 1$	54.85356* 11.22168*	43.63188* 11.22168*	Yes	2
No co-integration between market capitalization and oil	$r = 0$ $r \leq 1$	19.68516* 0.102925	19.58223* 0.102925	Yes	1
No co-integration between market capitalization and WPI	$r = 0$ $r \leq 1$	17.55103* 5.026420*	12.52461* 5.026420*	Yes	2

Source: Authors' computation.

Notes: *Significant at 5 per cent; **Significant at 10 per cent.

Sub-period Results

Period I: July 1997–July 2007 (121 months)

The results of granger causality test for sub-period I presented in Table 9 are significantly different from the results obtained in Table 10 for the entire study period. In sub-period I, only IIP is granger caused by Sensex and market capitalization and WPI is granger causing both Sensex returns and market turnover. There is bi-directional causality between Sensex & oil prices, WPI & market capitalization and IIP & WPI. Further, money supply is granger causing IIP and IIP is granger causing exchange rate.

Period II: August 2007–June 2011 (47 months)

From Table 9, it can be said that after August 2007, in the short run, the stock market performance cannot be predicted with the help of any one of the six macroeconomic variables considered. Rather, it is preceding the crucial macroeconomic variables, especially IIP, oil prices and interest rate. Also, bi-directional causality can be seen among WPI & oil prices and IIP & money supply.

Since causality results overall indicate that changes in stock market precedes the changes in economic variables, we can say that expectations about economic performance are first reflected in stock market indicators than in the real economic indicators. Hence, revival of investors' confidence is crucial for conducive economic environment and real economic recovery.

Johansen Co-integration Test

Johansen co-integration test is employed to examine the long-term causal relation between stock market performance and macroeconomic variables. We have performed both bi-variate and multi-variate Johansen's co-integration tests.

Bi-variate Johansen's Co-integration Results

Co-integration results as shown in Table 10 indicates that Sensex—IIP (at 10 per cent level of significance); Sensex—money supply and Sensex—oil prices (at 5 per cent level of significance) are cointegrated in the long run. In case of Sensex—exchange rate; Sensex—interest rate and Sensex—WPI, there is no evidence of co-integration. Further, there appears cointegration between market turnover—money supply; market turnover—oil prices; market capitalization—IIP; market capitalization—money supply; market capitalization—oil prices and market capitalization—WPI (at 5 per cent level of significance).

If the Johansen's co-integration test is applied on the *log series* of variables, only one pair is found to have cointegrating relationship, namely, market turnover and exchange rate.

Sub-period Results

The results of sub-periods are significantly different from the results obtained for the entire study period. While, during the entire study period, Sensex is having long-run relationship with IIP, money supply and oil prices, during sub-period I, Sensex is related to IIP and money supply and

during sub-period II, Sensex is co-integrated with only interest rate. Further, market turnover is cointegrated with money supply and oil prices during the entire study period, it is co-integrated with only money supply in sub-period I, and in sub-period II, it is co-integrated with none of the macroeconomic variables taken up in the present study. Similarly, market capitalization is co-integrated with IIP, money supply, oil prices and WPI during the entire period of study; it is co-integrated with all the six macroeconomic factors in sub-period I and with only interest rate and oil prices in sub-period II.

Multi-variate Johansen's Co-integration Results

Multi-variate Johansen's co-integration test results revealed that there are five cointegrating relationships in the long run among the stock market indicators and macroeconomic variables. The results of multi-variate Johansen's co-integration test on *log* series of the variables revealed the existence of two co-integrating relationships in the long run among Sensex and six macroeconomic variables taken up in the present study. Further, there are two co-integrating relationships found among market capitalization and macroeconomic variables. However, there is just one co-integrating relationship found among market turnover and macroeconomic variables.

Sub-period Results

During sub-period I, the results of multi co-integration test reveal the presence of three co-integrating relationships among Sensex and six macroeconomic variables taken together. market capitalization and macroeconomic variables have four co-integrating relationships in the long run. Further, there are only two co-integrating relationships among market turnover and macroeconomic variables. Results of sub-period I on *log* series of variables revealed the presence of seven cointegrating relationships among stock market indicators and macroeconomic variables.

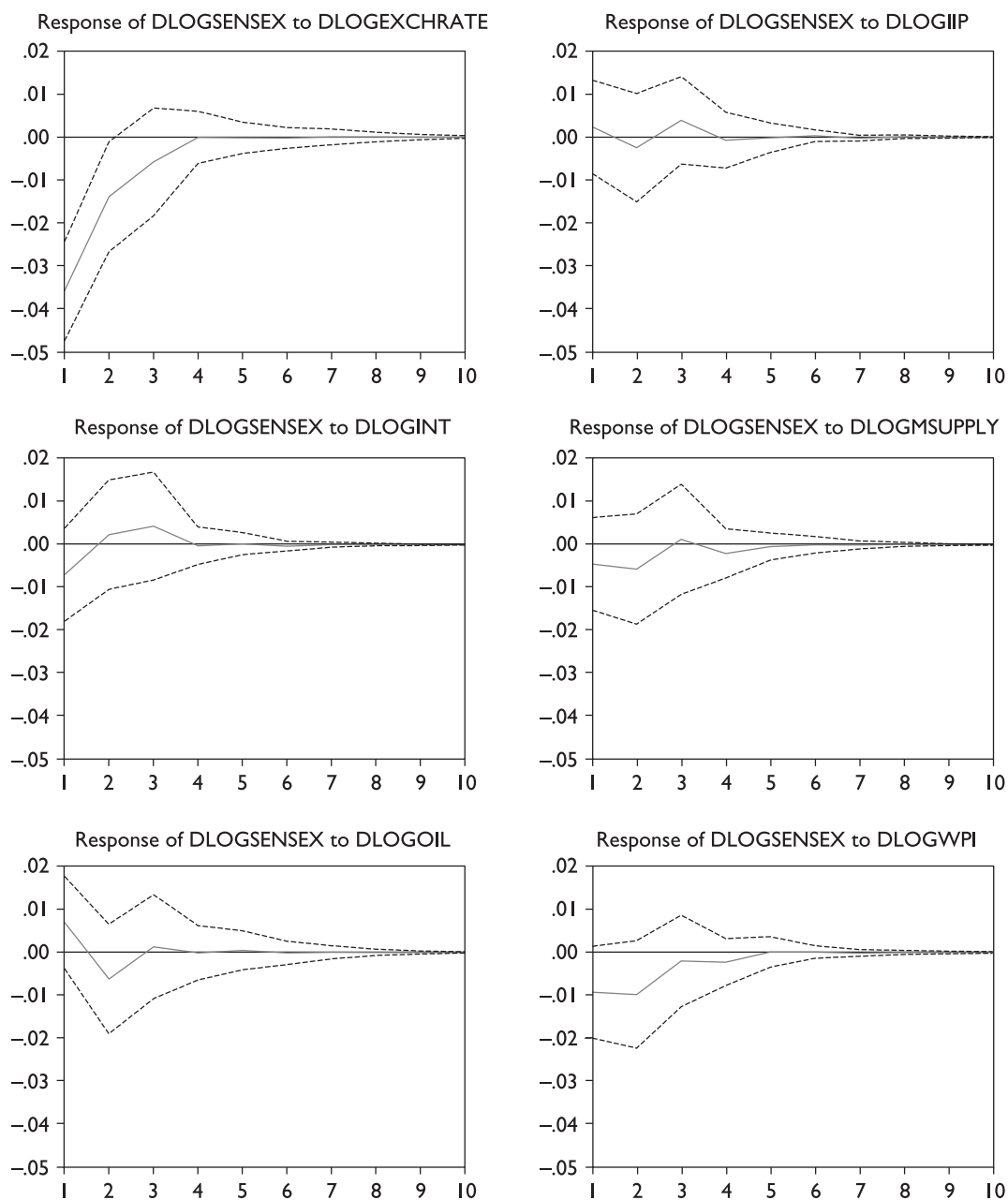
During sub-period II, there is just one co-integrating relationship in the long run among stock market indicators and macroeconomic variables. Results of sub-period II on *log* series of variables revealed the presence of two cointegrating relationships among stock market indicators and macroeconomic variables.

Impulse Response Analysis

Impulse Response Analysis has been performed to check the response of stock market indicators to shocks created in the real economy. Graph 1 shows that one standard deviation shock generated in exchange rate and inflation has a negative and significant effect on stock returns which lasts for about four to five months, while the response of Sensex returns to shocks created in IIP and oil prices is positive and significant. Similar results are obtained for the response of other stock market indicators to the shocks created in the real economy.

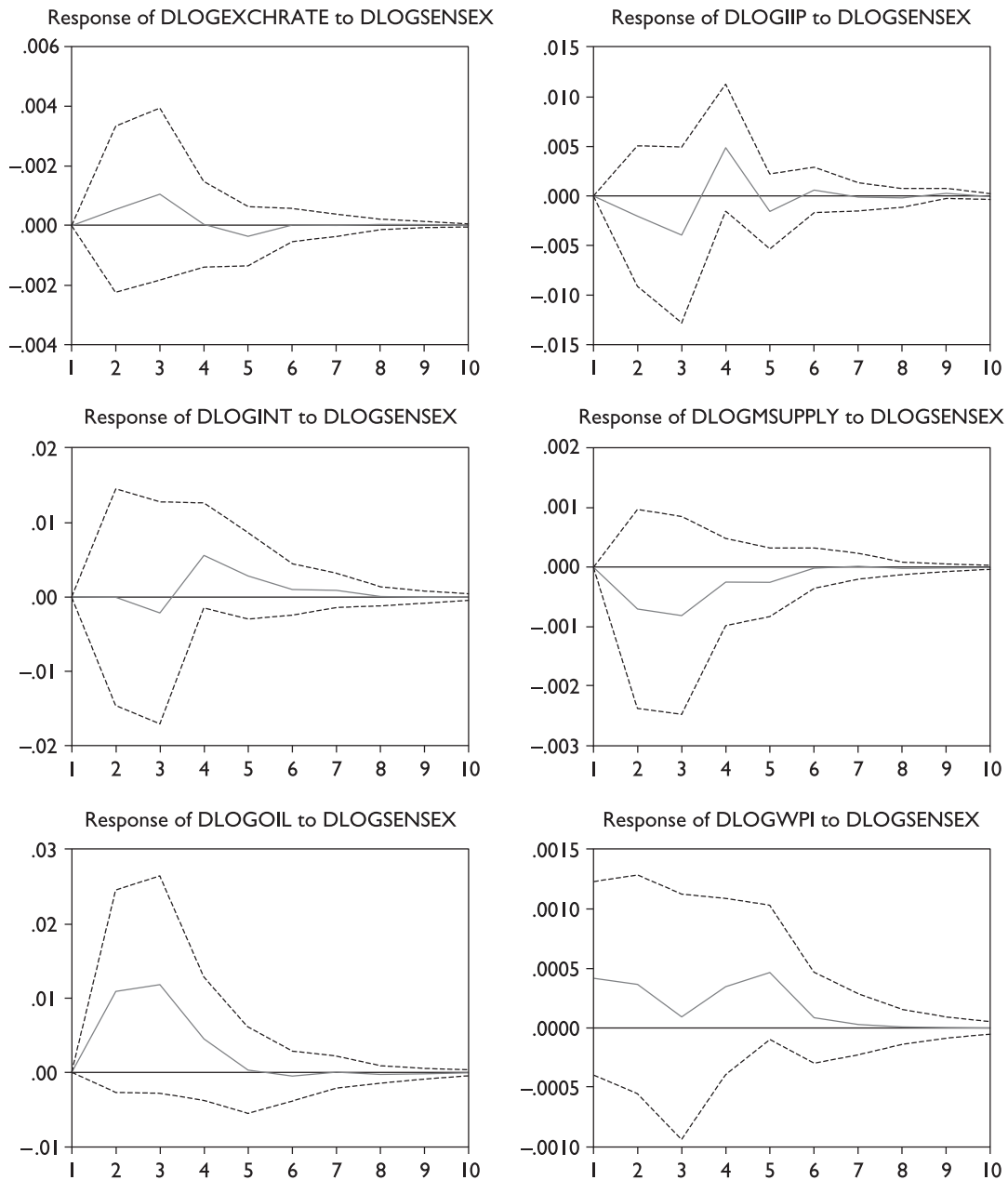
If we look at the response of macroeconomic variables to the shocks created in the stock market with the help of Graph 2, it can be observed that there is significant positive response of oil prices, exchange rate and inflation to one standard deviation shock created in the sensex returns. Further, the response of IIP and money supply is negative and significant.

Response to Cholesky One S.D. Innovations ± 2 S.E.



Graph 1. Impulse Response Analysis Results

Source: Authors' computation.

Response to Cholesky One S.D. Innovations ± 2 S.E.**Graph 2.** Impulse Response Analysis Results

Source: Authors' computation.

Conclusion

This article examines the causal relation between stock market performance and macroeconomic variables, namely exchange rate (₹ per \$), IIP WPI, oil prices (\$ per Barrel), money supply and interest rate (91 days T-Bills implied yield) using monthly data that span from July 1997 to June 2011 by means of descriptive statistics, correlation analysis, Factor analysis, regression analysis and some sophisticated statistical techniques such as Unit root test, ARCH test, Granger causality test to check for the short-run causal relation, Johansen co-integration test to determine the long-run relation and Impulse Response Analysis has also been performed to check the response of stock market indicators to shocks created in the real economy with the help of graphs. Monthly index values of BSE Sensex are used as a proxy for aggregate stock returns. Further, BSE market capitalization and market turnover are used to study the overall stock market performance. The whole study period is divided into two sub-periods with the break taking place in August 2007 that was the start of global financial crisis and all the tests have been applied for the entire period as well as two sub-periods.

The results are mixed but interesting. While, there is significant correlation among stock market variables and macroeconomic factors except for exchange rate, the Granger causality is prevalent only in respect of few variables out of which exchange rate is the single variable that is granger causing BSE Sensex and BSE India market Capitalization. money supply is also granger causing market turnover. IIP, oil prices and WPI are granger caused by stock market indicators. All this shows that much of the variation in economic variables is caused by stock market indicators and not the vice versa. The results of Granger causality also vary during the sub-periods. During sub-period I, only IIP is granger caused by Sensex and market Capitalization and WPI is granger causing both Sensex returns and market turnover. Further, there is bi-directional causality between Sensex & oil prices and WPI & market capitalization. During sub-period II, the stock market performance cannot be predicted with the help of any one of the six macroeconomic variables considered. Rather, it is preceding the crucial macroeconomic variables, especially IIP, oil prices and interest rate. Before applying Granger causality and regression tests, the time series were tested for stationarity using ADF and PP unit root test.

The factor analysis left us with three major factors which we named as inflation, exchange rate and interest rate. The results of multi-variate regression analysis reveal that the overall explanatory power of the model is 23.8 per cent, 23.3 per cent and 16.9 per cent respectively for Sensex, market capitalization and market turnover. The comparison of results of two sub-periods depicts that the explanatory power of the model significantly increases in sub-period II as compared to sub-period I. Hence, the relationship has become stronger in the recent time period. The application of ARCH technique on our regression model reveals that there is no problem of heteroskedasticity in case of Sensex and market capitalization as dependent variables.

The results of Johansen's co-integration test reveal that there is long-run causal relation between stock market performance and oil; stock market performance and money supply; Sensex and IIP; market capitalization and IIP; market capitalization and WPI which implies select macroeconomic factors have direct long-run equilibrium relationship with the stock market performance indicators. However, sub-period analysis revealed reduced cointegration in recent time period. Moreover, as compared to developed markets, the relationship is weaker in India. The study clearly indicates that it is not the real economic variables that precede stock market movements but, stock market performance precedes the real economy more. These findings are reinforced by the graphs generated by impulse response analysis. One plausible reason may be that stock markets reflect the sentiments as well the expected moves in real activity.

However, there are few long-run relationships among the stock market indicators and macroeconomic factors as depicted by cointegration test.

Policy Implications

The research findings have important implications for market regulators, companies and investing community. They are important for the market regulators because there is long-term co-integration found between stock market variables and some of the macroeconomic variables. Although there is no serious cause of worry, there is still room for improvement as Sensex is not found to be co-integrated with interest rate, inflation and exchange rate. So, market regulators can look for reforms in these areas. This also puts forward a case for de-regularizing the interest rates. If we talk about investors, they are more interested in investment strategies that can outperform the market and make them earn supernormal profits. But, our results indicate that the macroeconomic variables cannot be used to devise profit earning strategies from the stock market, because it follows stock market fluctuations rather than precedes the stock market. So, investors should not heavily rely on past macroeconomic data. Further, if we distinguish between small investors and large institutional investors like FIIs, large investors still have the benefit of international diversification which is not available to small investors, the less informed section of investing community.

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